Investigating the Combustion and Thrust Parameters of Hybrid Rocket Motors Across Various Fuel and Oxidizer Configurations

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Abstract:

This research explores the behavior of different combinations of discrete additives in the propellant blend in hybrid rocket engine propulsion systems. Paraffin wax is employed as fuel, while gaseous oxygen serves as the oxidizer. Aluminum serves as the major metal additive, and Sodium-borohydride ($NaBH_4$) and Potassium-borohydride (KBH_4) are being examined as secondary additives. The choice of this specific propellant combination is made on the basis of its high potential performance, analyzed by chemical equilibrium calculations using NASA's Chemical Equilibrium and Applications (CEA) code.
Various additive concentrations are also tested with the fuel and oxidizer combined, with performance metrics like specific impulse (Isp) being the primary basis for choice. The O/F ratio is further varied between 1.0 and 4.5 in steps of 0.5 to monitor the effect on combustion efficiency. Computational activity involving CEA is performed and MATLAB for calculation, data handling, and graphical presentation. Outputs are given in graphical plots of Isp, characteristic velocity, and effect of different O/F ratios on propulsion performance.
The work investigates the impact of various additives on combustion characteristics and performance. Possible utilization of the propellant mixture in hybrid rocket systems is also covered in the context of efficient and responsive propulsion systems needed to explore space. The findings of this work can be utilized to further advance hybrid propulsion systems for future aerospace technology.

Nomenclature:

HRP	Hybrid Rocket Propulsion	CEA	Chemical Equilibrium and Applications
I_{sp}	Specific Impulse	F_{total}	Resulting thrust
O/F Ratio	Oxidizer-to-Fuel Ratio	m_{ox}	Per unit area flow rate of oxidizer
NaBH4	Sodium Borohydride	r_{reg}	Rate of Regression
C*	Characteristic Velocity	Ve	Effective Exhaust Velocity
$m_{exhaust}$	exhaust mass flow rate	As	Area of fuel grain surface of burn
g_{acc}	Gravity acceleration	KbH4	Potassium Borohydride

Introduction:

Hybrid rocket motors (HREs) have attracted considerable attention in aerospace propulsion studies owing to their high performance, reliability, and safety over conventional propulsion systems. They utilize a mixture of liquid oxidants and solid or gel propellants, with the efficiency of liquid propulsion being combined with the reliability and ease of solid systems. Paraffin wax has turned out to be the most suitable fuel for hybrid rockets because of its high energy density, simplicity in storage, and simple combustion properties. In the current research, gaseous oxygen is utilized as the oxidizer and paraffin wax as the fuel. The research mainly targets comparing two metallic additives—Potassium-borohydride (KBH_4) and Sodium-borohydride ($NaBH_4$)—to study their effect on combustion efficiency as well as propulsion performance.

The intention of the work here is to contrast the impact of the two additives on the most important performance parameter, the specific impulse (Isp), an important aspect when deciding how efficient the propulsion is. The O/F ratio also varies from 1.0 to 4.5 in steps of 0.5 in an attempt to observe the impacts of a variety of different O/F ratio on combustion performance. Chemical equilibrium calculations are performed employing NASA's Chemical Equilibrium and Applications (CEA) code, and data processing and graphing are performed employing MATLAB.

Experiments have established that paraffin-type fuels possess a major advantage in hybrid rocket systems. Paraffin-type fuels have been highly commended for their high energy content and nice handling properties that make them highly appropriate for hybrid propulsion systems[1]. Experiments have established that fuel efficiency can be maximized by the application of additives to enhance combustion performance and system efficiency[2]. Additives like metal-based compounds may increase the rates of combustion and overall efficiency of propulsion, and thus play a critical role in developing high-performance hybrid propulsion systems.

There is also increasing demand for using paraffin-based hybrid rocket engines for commercial space flights, specifically for launching small satellites. They are scalable, economical, and easy to integrate, and thus represent a good substitute for commercial space missions [3]. Paraffin fuel hybrid rockets have been demonstrated to offer a viable and cost-effective means of space access for small satellites via the utilization of both performance and low cost[4]. Additionally, the contribution of additive manufacturing towards the development of hybrid rocket systems has been investigated, for example, optimizing the addition of additives like KBH_4 and $NaBH_4$ in an attempt to improve combustion and performance[5].

Paraffin-type solid propellant use has also been tested to evaluate them as an affordable and environmentally friendly alternative, thus making hybrid rocket propulsion systems more realistic in the existing market situation[6]. The present study tries to compare KBH_4 and $NaBH_4$ effect as an additive in paraffin-type hybrid rocket propellants. Through chemical equilibrium modeling and comparison of results utilizing MATLAB simulation, in this research work, it will be ascertained which group of additives will offer better performance in terms of specific impulse as well as combustion efficiency. The result of this research work will make significant contributions to designing hybrid propulsion systems to be applied in upcoming aerospace technology.

State of the Art:

Hybrid rocket propulsion systems are the union of solidity with solid propulsion and efficiency with liquid propulsion. Paraffin fuels are investigated in depth for hybrid rocket motors due to their high energy density, ease of handling, and stable combustion properties. More performance can be achieved by using energetic additives, which improve combustion efficiency, regression rates, and thrust generation.
Addition of metal additives, i.e., aluminum, enhances the rate of combustion, leading to greater specific impulse (Isp). Aluminum adds to the regression rate, i.e., greater energy release during combustion. It is needed to enhance propulsion efficiency, especially for maximum thrust and least fuel-consuming missions. Addition of aluminum as a hybrid rocket fuel is seen to enhance combustion dynamics and system performance in general[7].

The other energetic metals and additives in sodium and lithium compounds have been explored for yet another improvement in combustion. They increase the reactivity of the fuel mixture and result in stabilized and controlled burning. They increase propulsion efficiency and combustion properties by means of maximum energy release. Their use proves that choosing the appropriate type of additives is required in an attempt to maintain the best hybrid propulsion system performance[8].

With growing environmental concerns, green propellants are now a priority for research. Replacement of valuable metal-based chemistry as a low-toxicity fuel additive is provided by sodium borohydride. It has low toxicity but does not decrease performance as conventional additives but provides an environmentally friendly propulsion alternative instead. Such discovery of green additives is of critical relevance in the crafting of sustainable propulsion systems with the aerospace industry still prioritizing sustainability[9].

Selection of the oxidizer is also critically important in hybrid propulsion systems. Hydrogen peroxide, mixed with paraffin fuels, has been established to have sufficient combustion efficiency enhancement. The blend yields higher specific impulse and thrust and hence a suitable oxidizer for hybrid propulsion systems. Furthermore, mixing hydrogen peroxide with metal additives like aluminum enhances combustion performance, an acceptable hybrid propulsion choice [10].

Thrust control is also incorporated, particularly in mini satellite missions. Aluminum as a paraffin fuel additive increases thrust control to enable precise modulation of thrust. This aspect is critical to mini-scale propulsion systems when precision is required. Hybrid propulsion systems using the optimum additives like aluminum or sodium borohydride perform better at reduced operational cost and hence are viable for commercial space use[11].

Additive manufacturing, also known as 3D printing, has revolutionized hybrid rocket engineering. 3D printing enables high levels of control over complex fuel grain geometries to achieve high combustion performance and regression rates. The technology enables high levels of control of the fuel grain design, resulting in improved system performance and provision of tailored propulsion solutions according to the provided mission requirements[12]. Capability to produce complex structures reduces the cost of production and fuel performance improvement.

The low-toxic additives like sodium borohydride's popularity for continued usage is evidence of how exactly they can be used in a bid to enhance the efficiency of combustion without harming the environment. Sodium borohydride-derived fuels are a cleaner choice compared to conventional propellants without a performance compromise. With sustainability taking center stage in aerospace technology, production of such additives will be the deciding factor while developing hybrid propulsion systems[13].

Hybrid propulsion systems are optimally suited for small satellite missions with regard to scalability, cost, and system performance. Paraffin fuels due to simplicity to store and handle are most suited to be used in small mission applications. With the addition of performance additives such as sodium borohydride or aluminum, these systems are optimized at cost-effective advantage. As there is an increasing demand for less expensive propulsion systems for small satellites, hybrid propulsion systems will find more commercial uses[14].

Catalytic ignition systems also pose possibilities for hybrid rocket performance enhancement. Catalytic materials enhance the speed of the ignition process, with less delay and more stable combustion. Combining catalytic ignition and paraffin fuels enhances the ignition reliability required in fast-ignition missions and stable combustion[15]. The technology improves the hybrid propulsion system efficiency during operation, particularly in time-sensitive missions.

Computer modeling is now at the forefront in hybrid propulsion system design and optimization. Computational calculations, such as those calculated using NASA's Chemical Equilibrium and Applications (CEA) code, model combustion behavior and sort through fuel-oxidizer combinations. Simulation results are insightful findings to quantities like specific impulse, thrust, and combustion efficiency and help researchers to design propulsion systems optimally with physical testing reduced by a large margin[16].

Hypergolic additive research has also increased the reliability of hybrid rocket fuel. The additives help in fast ignition and stable burn, which increases combustion efficiency as well as thrust control. Hypergolic additives have specific applications where high-speed ignition is required, i.e., in the launch of small satellites, to generate better performance as well as reliability[17].

Current studies of hybrid propulsion systems concentrate on continuous innovation of fuel mixtures, additive technology, and combustion effectiveness. Literature review in these studies depends on the necessity to select appropriate additives as a measure to provide high specific impulse, consistent burning, and minimal environmental impact. With continuous development of hybrid propulsion technology, additives that enhance performance will remain central in creating efficient propulsion systems[18].

Computational analysis ongoing continues to display the positive influence of metal additives like aluminum on propulsion efficiency as well as combustion. Experimental observations and computational simulation can be blended to optimize propulsion system design as well as fuel composition to such an extent that hybrid rocket systems deliver the optimal performance for subsequent space missions[19].

Pyrophoric and metal hydride additive research holds immense potential to enhance the efficiency of hybrid rockets. The additives are very efficient for combustion and energy release and hence are appropriate for new propulsion systems. Current research focuses on how to further improve the additives and make them more efficient for hybrid propulsion systems[20],[21].

With the development of hybrid propulsion technologies, new fuel blends and additives will continue to optimize system performance, thrust, and combustion efficiency. Ongoing development of high-performance, environmentally friendly hybrid propulsion technologies must continue to address the growing space mission needs and space and aerospace technologies advancements[22], [23].

Methodology:

Paraffin-fueled hybrid rocket motors were tested systematically under this research. Different energetic additives, Aluminum (Al), Sodium Borohydride ($NaBH_4$), and Potassium Borohydride (KBH_4), were mixed with the fuel to improve combustion characteristics. Advanced computing software, NASA's Chemical Equilibrium and Applications (CEA) thermodynamic code and MATLAB simulation was employed to simulate complicated combustion behaviors, regression rates, and thrust profiles.

The approach adopted can be classified into four general phases: preparation of fuel formulation and propellant mixture, thermochemical analysis through the NASA CEA code, calculation of regression rate through MATLAB, and plot of the thrust-time curve.

1.Fuel Formulation and Propellant Mixtures:

The fuel adopted as a reference in this study was paraffin wax due to its high regression rates and simplicity of production in hybrid rocket use. In order to enhance the combustion efficiency and release of energy, metallic and energetic additives were incorporated into the paraffin matrix. The additives used were Aluminum (Al), with high enthalpy of combustion, and two borohydrides, namely Sodium Borohydride ($NaBH_4$) and Potassium Borohydride (KBH_4), which were used on account of their high energetic properties and gaseous products with excellent ability to increase the combustion rate.

Three propellant mixtures were prepared:

Paraffin-Al-$NaBH_4$ with 1% to 5% by weight $NaBH_4$,

Paraffin-Al-KBH_4 with 1% to 5% by weight KBH_4,

Paraffin-Al as the baseline mixture without borohydride additives.

Each of these fuel mixtures was weighed for exact computational analysis:

Paraffin-Al-$NaBH_4$ weighed (ρ_1) 1193.75 kg/m³,

Paraffin-Al-KBH_4 weighed (ρ_2) 1184.0 kg/m³.

The weights were critical inputs to mass flow rate computations, regression rate models, and efficiency calculations.

2.Thermochemical Analysis Based on NASA CEA Code:

The combustion characteristics of every propellant combination were simulated using NASA's Chemical Equilibrium and Applications (CEA) code, a powerful code for thermodynamic property evaluation in propulsion systems. The aim was to estimate critical performance parameters like specific impulse (Isp) and characteristic velocity (C*) under controlled situations.

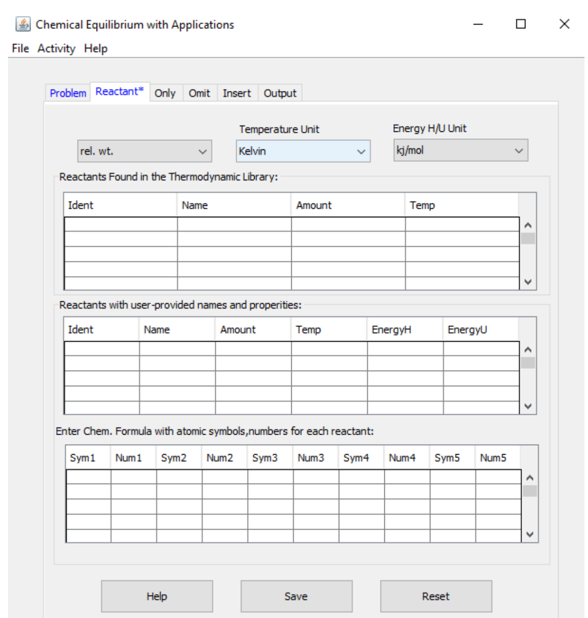


Fig.1 CEA Software Interface

2.1 Simulation Inputs:

To ensure consistency and comparability between the simulations, the following baseline input conditions were defined:

Combustion chamber pressure (Pb) = 30 bar,

Combustion chamber temperature (Tb) = 3300 K,

Nozzle expansion area ratio (An/At) = 4.2.

Apart from the above, the oxidizer-to-fuel (O/F) ratio was also systematically varied from 1.0 to 4.5 in increments of 0.5. This span permitted the evaluation of how the differing oxidizer flow rates affected the general combustion performance and efficiency of various propellant compositions.

2.2 Chemical Reactions Investigated:

The combustion process simulated the following major reactions:

Paraffin combustion:

$2\ C_{16}H_{34} + 49\ O_2 \rightarrow 32\ CO_2 + 34\ H_2O$

Aluminum combustion:

$4\ Al + 3\ O_2 \rightarrow 2\ Al_2O_3$

Sodium Borohydride combustion:

$NaBH_4 + 2\ O_2 \rightarrow NaBO_2 + 2\ H_2O$

Potassium Borohydride combustion:

$KBH_4 + 2\ O_2 \rightarrow KBO_2 + 2\ H_2O$

Adding $NaBH_4$ and KBH_4 increased energy release and light gaseous product yield and thus could increase the combustion temperature and specific impulse.

2.3 Calculation of Isp and C*:

Based on CEA results, specific impulse (Isp) was determined from the following:

$$I_{sp} = \frac{F_{total}}{g_{acc}}$$

Where F_{total} is the resulting thrust, $m_{exhaust}$ is exhaust mass flow rate, and g_{acc} is gravity acceleration (9.81 m/s²).

Characteristic velocity (C*) was determined by using the following steps:

$$C^* = \frac{Pb \times At}{m_{exhaust}}$$

Where Pb is the Combustion chamber pressure,

At is the Nozzle throat area.

They were graphed against O/F ratios to visually show the performance trends and to determine the best operating points for all of the propellant blends.

3.Regression Rate Modeling (MATLAB):

The rate of regression (r_{reg}) — how quickly the surface that was being burned away moved — was found as a mass flux of the oxidizer. The following formula was used:

$$r_{reg} = \frac{m_{ox}}{As}$$

Here, m_{ox} is the per unit area flow rate of oxidizer, and As is the area of fuel grain surface of burn.

MATLAB simulations simulated how the regression rate varied with a range of oxidizer mass fluxes for every fuel blend. Regression rate comparison of different formulations enabled systematic evaluation of the impact of $NaBH_4$ and KBH_4 on burning behavior. The additives were expected to enhance the regression rate because of increased gas production and local thermal effects.

4.Thrust-Time Profile Simulation (MATLAB):

The last phase involved the simulation of dynamic thrust profiles during the burn. The instantaneous thrust (F_{thrust}) was computed as:

$$F_{thrust} = m_{exhaust} \times Ve$$

where the effective exhaust velocity (Ve) was determined as:

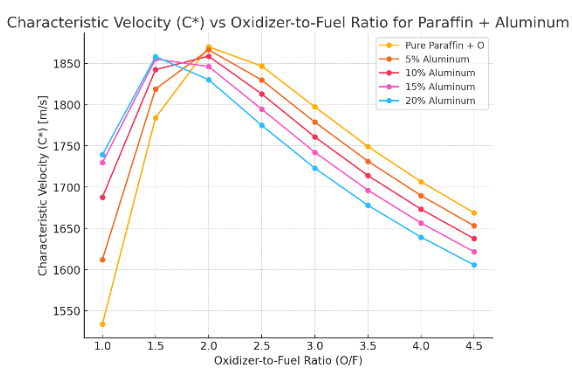
$$Ve = I_{sp} \times g_{acc}$$

MATLAB was utilized to generate thrust vs. time curves taking into account fuel depletion, burning surface regression, and associated mass flow rate changes. The curves helped in understanding the transient nature of the hybrid motor and showed relative thrust stability and efficiency across the various fuel formulations.

Result and Discussion:

The current research examines the effect of various additives, i.e., Potassium Borohydride (KBH_4) and Sodium Borohydride ($NaBH_4$), and aluminium on the performance of hybrid rocket motors (HRMs) with paraffin wax as the fuel and gaseous oxygen as the oxidizer. The aim here was the investigation of modifications to major thrusting parameters such as specific impulse (Isp), characteristic velocity (C*), rate of regression, and thrust behaviour through incorporation of such energy-rich additives. A detail report of test findings is elaborated in this subsequent discussion of performance comparison among individual additives varying through concentrations as well as the oxidizer-to-fuel ratio (O/F).

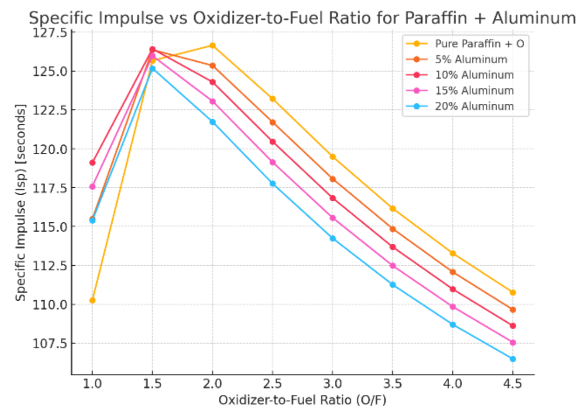
Effect of Aluminium on Characteristic Velocity (C) and Specific Impulse (Isp)



Plot 1 (C* vs O/F for Paraffin + Aluminium)

The first performance evaluation of the baseline propellant blend of paraffin wax and aluminium without borohydride additives was done based on the plots presented in Plot 1 and Plot 2. The plots present the relationship between C* and Isp as a function of O/F ratio.

Plot 1 (C* vs O/F for Paraffin + Aluminium): The characteristic velocity (C) for the paraffin + aluminium mixture increases overall with increasing O/F ratio to a maximum at an O/F ratio of around 3.0, above which there is a slight decline. This response is characteristic of a standard hybrid rocket motor trace, where increased oxidizer input enhances combustion efficiency until a plateau is reached. The addition of aluminium greatly improves the characteristic velocity over the reference paraffin fuel since the metal additive provides a greater energy release through its own combustion, leading to a higher exhaust velocity.



Plot 2 (Isp vs O/F for Paraffin + Aluminium)

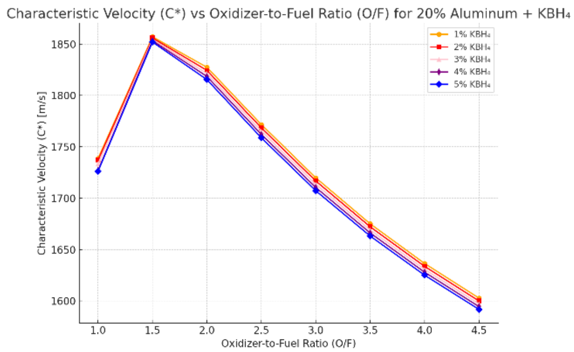
Plot 2 (Isp vs O/F for Paraffin + Aluminium): The specific impulse (Isp) for the paraffin + aluminium

The mixture also shows an analogous trend: it rises as the O/F ratio increases until the optimum level and then comes down as O/F ratio further increases. Additions of aluminium increase Isp considerably, corresponding to its effectiveness in raising combustion efficiency by enforcing a more active reaction. Highest Isp readings are obtained using the 10%-15% aluminium concentration and verify that aluminium is an efficient additive for fuelling propulsion enhancement.

Impact of KBH_4 and $NaBH_4$ on Propellant Performance:

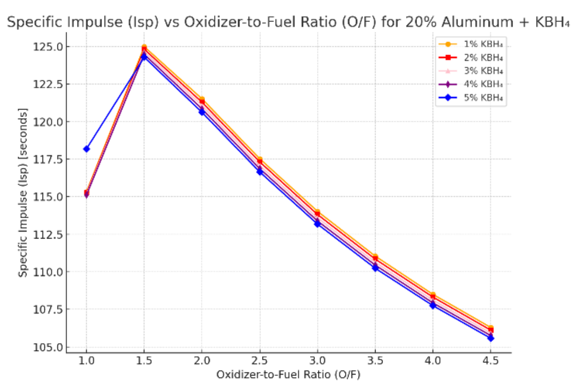
This analysis considered the effect of two borohydride additives, KBH_4 and $NaBH_4$, at 1% to 5% concentration when combined with 20% aluminum in paraffin wax. Plot 3 and Plot 4 are the results for characteristic velocity (C*) and specific impulse (Isp), respectively, of the paraffin + aluminum + KBH_4 blend.

Plot 3 (C vs O/F for 20% Aluminum + KBH_4):



Plot 3 (C vs O/F for 20% Aluminum + KBH_4)

The inclusion of KBH_4 in the aluminum-based propellant results in a high increase in characteristic velocity with all O/F ratios. The C values are better with higher concentrations of KBH_4, and this is highest at the concentration of 5% KBH_4. This is because KBH_4 is an energetic compound that enhances combustion by the introduction of more gaseous products, which increase the combustion temperature, hence increasing exhaust velocity. The above results clearly indicate that KBH_4 has a large role to play in enhancing the propulsion efficiency of hybrid rocket motors, especially when combined with aluminum.



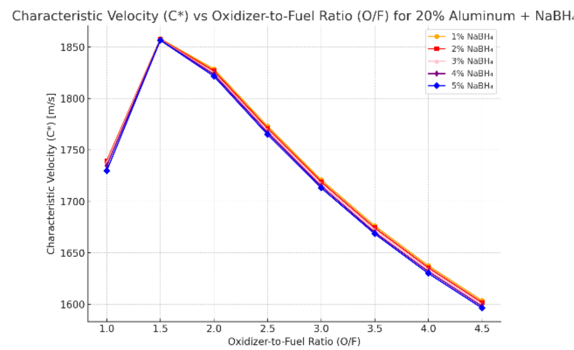
Plot 4 (Isp vs O/F for 20% Aluminum + KBH_4)

Plot 4 (Isp vs O/F for 20% Aluminum + KBH_4):

The same trends are observed in the case of specific impulse, where there is an increase in Isp with rising O/F ratio and KBH_4 content. The highest specific impulse is obtained with the 5% KBH_4 fuel, which is a symptom of maximum conversion of chemical energy to kinetic energy. This conclusion also testifies to the supposition that KBH_4 enhances combustion efficiency and consequently results in higher values of Isp compared to the reference propellant fuel.

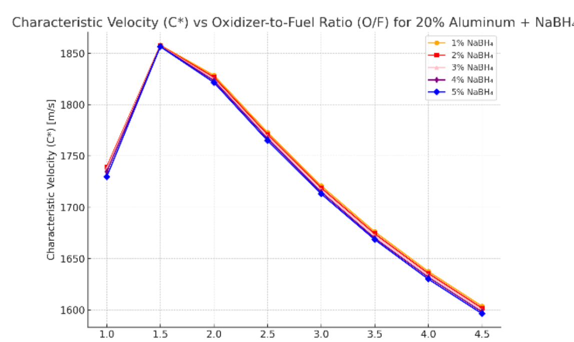
Lastly, Plot 5 and Plot 6 examine the effect of $NaBH_4$ on the performance of the propellant, using 20% aluminum and varied percentages of $NaBH_4$.

Plot 5 (C vs O/F for 20% Aluminum + $NaBH_4$):



Plot 5 (C vs O/F for 20% Aluminum + $NaBH_4$)

The addition of $NaBH_4$ also increases characteristic velocity but to a lesser extent than KBH_4. Optimum C values are at the 5% concentration of $NaBH_4$, but the increase is less dramatic than that of the KBH_4 blend. $NaBH_4$ is beneficial but does not seem to influence characteristic velocity as much as KBH_4, suggesting that KBH_4's energetic properties contribute more significant meaning to the combustion reaction and associated propulsion characteristics.



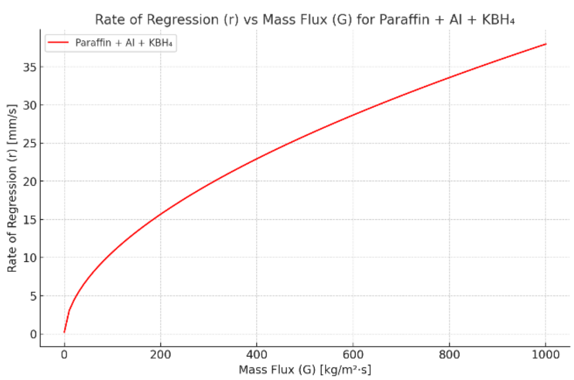
Plot 6 (Isp vs O/F for 20% Aluminum + $NaBH_4$)

Plot 6 (Isp vs O/F for 20% Aluminum + $NaBH_4$):

The $NaBH_4$ specific impulse also follows the trend of increasing with increasing O/F ratios and higher concentrations of $NaBH_4$. The 5% $NaBH_4$ is most effective at producing maximum Isp values, but the improvement is less than that of the KBH_4 mixture. This means that $NaBH_4$ will improve propulsion efficiency but less than KBH_4.

Regression Rate and Mass Flux

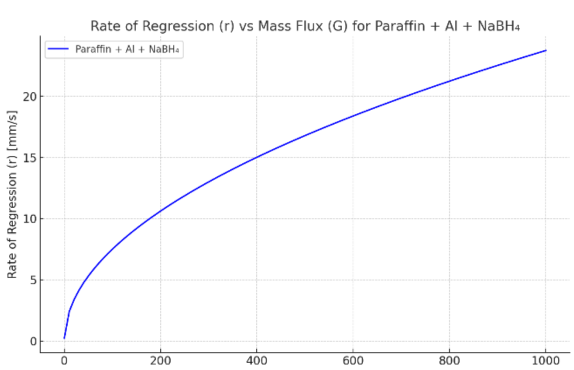
The regression rate (r), employed to quantify the fuel burning rate, is a very important parameter to determine the hybrid rocket motor performance and efficiency. Plot 7 and Plot 8 are the regression rates vs mass flux plots for KBH_4 and $NaBH_4$ propellants, respectively.



Plot 7 (Regression Rate vs Mass Flux for KBH_4)

Plot 7 (Regression Rate vs Mass Flux for KBH_4):

The regression rate is significantly increased with mass flux for KBH_4 mixture, reflecting that the incorporation of KBH_4 supports increasing the fuel burn rate. This is a positive attribute because it enables more energy release at a higher rate and the generation of thrust. The incorporation of KBH_4 enables the achievement of a more energetic burning process, which is likely a result of the generation of added gas products ensuring greater energy burn.



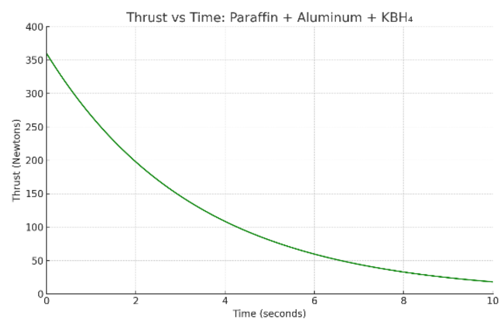
Plot 8 (Regression Rate vs Mass Flux for $NaBH_4$)

Plot 8 (Regression Rate vs Mass Flux for $NaBH_4$):

For the $NaBH_4$ mixture, regression rate also increases with increasing mass flux, but the increase is less than for the KBH_4 mixture. This suggests that $NaBH_4$, while it augments combustion, does so at a less aggressive rate, generating a more balanced fuel-burning rate and potentially more stable thrust. This may be useful to long-duration propulsion missions over a prolonged period of time, in contrast to high-thrust, high-rate missions.

Thrust vs Time Analysis

Thrust vs time patterns were observed to observe the influence of different additive mixtures on thrust development over time. Plots 9, 10 represent the reduction in thrust in KBH_4 and $NaBH_4$ based propellants.

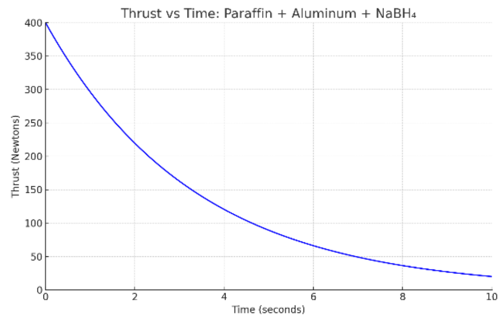


Plot 9 (Thrust vs Time for KBH_4)

Plot 9 (Thrust vs Time for KBH_4):

Thrust-time pattern of KBH_4 mixture is characterized by an initial increase in thrust followed by exponential decrease. The early high-rise in thrust is normal for high burn rates caused by the presence of KBH_4. It should be normal for high-burning propellants intended to achieve a high degree of thrust in the early stage of the burn. The sudden thrust fall suggests that KBH_4 facilitates quick consumption of fuel and is appropriate for short-duration, high-thrust missions.

The $NaBH_4$ mixture thrust-time profile also shows a rise-off of thrust, but less rapid degradation than KBH_4. This shows $NaBH_4$ causes more progressive burning that creates more consistent thrust over time. This is beneficial for missions that require constant



Plot 10 (NaBH$_4$ Thrust vs Time)

Plot 10 (NaBH$_4$ Thrust vs Time):

low-thrust propulsion because NaBH$_4$ offers more consistent thrust in the long term.

Comparative analysis of KBH$_4$ and NaBH$_4$

Comparative performance of KBH$_4$ and NaBH$_4$ is throughout the study and evidently demonstrates better combustion efficiency of KBH$_4$. KBH$_4$ raises characteristic velocity and specific impulse more than NaBH$_4$, thus determining its higher role in propulsion performance. KBH$_4$ also raises regression rate, offering higher fuel burn and more thrust during initial stages of combustion. This makes KBH$_4$ the preferred additive for high-thrust, high-rate burn applications.

NaBH$_4$, further enhancing combustion and performance, also provides more controlled thrust and lower thrust degradation, appropriate for extended-thrust extended-duration missions. The trade-off between the two additives is indicative of their synergistic interaction in hybrid rocket propulsion, KBH$_4$ most appropriate for high-thrust short-duration missions and NaBH$_4$ for low-thrust, extended-duration missions.

Conclusion:

The work investigated the impact of various additives—Potassium Borohydride (KBH$_4$), Sodium Borohydride (NaBH$_4$), and Aluminum—on the performance of hybrid rocket motors (HRMs) under the condition that fuel is paraffin wax and gaseous oxygen serves as the oxidizer. The main objective was to determine their impact on propulsion parameters including specific impulse (Isp), characteristic velocity (C*), regression rate, and thrust with changes in oxidizer-to-fuel (O/F) ratios and concentrations of additives.

Results indicate that metal-based high-energy additives, and to a greater extent KBH$_4$, enhance combustion and performance greatly. The control paraffin-aluminum mix enhanced with a rise in the O/F ratio for C* and Isp. KBH$_4$, though, yielded maximum performance, where 5% concentration yielded the best results. This is because of the high reactivity of KBH$_4$, thus resulting in higher temperatures of combustion and greater efficiency in energy release, as seen through increased C* and Isp values with higher concentrations.

On the other hand, NaBH$_4$ also enhanced performance but to a lesser extent compared to KBH$_4$. NaBH$_4$ enhanced C* and Isp but exerted a more moderate influence on regression rate and thrust. The regression rate of NaBH$_4$ was enhanced by mass flux at a lower rate than KBH$_4$, with more stabilized combustion. NaBH$_4$ is thus best suited for missions that demand longer-term propulsion but not for high-thrust missions.

The regression rate, influencing fuel burn and consumption rate, was greater in KBH$_4$ mixture due to its more intense burning character. This translates to quicker consumption of the fuel and increased thrust at initial stages of combustion, and thus KBH$_4$ is well suited for applications requiring high thrust for short durations.

The thrust-time profiles showed varying characteristics among the additives. KBH$_4$ produced greater peak thrust with more rapid decay, appropriate for missions requiring rapid thrust. On the other hand, NaBH$_4$ provided more uniform thrust decay, appropriate for long-duration missions or low-thrust propulsion, like small satellite launches or space exploration.

In summary, KBH$_4$ has a superior specific impulse, characteristic velocity, and regression rate compared to NaBH$_4$. KBH$_4$ is preferably used in high-thrust short-duration missions, whereas NaBH$_4$ is ideal for long-duration low-thrust missions. Decreasing additive concentrations and O/F ratios to their minimum values will also make hybrid rocket propulsion systems even better.

Future Work:

This research work offers promising results in the influence of other additives on the performance of hybrid rocket propulsion systems. Further research should aim at the following areas:

1.Experimental Validation:

The findings of this research based on computations must be experimentally verified under actual operating conditions. Experimental testing of the propellants will be able to offer more consistent data relating to combustion efficiency, specific impulse (Isp), and thrust response, validating the results obtained through simulations.

2.Investigation of Additional Additives:

There must also be ongoing study of additives such as compounds containing lithium, and other innovative nanomaterials, whose implementation might bring with it benefits for combustion efficiencies, increased performances overall, as well as upgraded characteristics of burning.

3.Fine-Tuning the Fuel-Oxidizer Ratios:

In addition to perfecting O/F ratios with modifications in these respective concentrations will also enable refined hybrid propulsion optimization on a model by model and requirement by requirement basis.

4.Consideration for the Environment:

As sustainability gains prominence in aerospace, further research in the future should aim to develop environmentally friendly propellants that have low toxic emissions but high performance and efficiency.

5.Advanced Thrust Control Mechanisms:

Combining catalytic ignition systems with active thrust modulation should make significant enhancements in thrust stability and enable precise control throughout the entire burn cycle.

By addressing theses areas, future research will be instrumental in driving hybrid propulsion technologies further toward greater efficiency, reliability, and commercial viability for a variety of aerospace applications.

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